

What we're doing

Implementing ``contacts-and-distances-v1`` as a Marifold first-class document structure: a set of plain modules + a ``cli.py`` driver that uses the shared toolkit in ``marifold_document_structures``. Layout: ``vocab.py``, ``parse.py`` (gemmi-backed PDB/mmCIF reader), ``generate.py`` (doc generation), ``inference.py`` (predict + evaluate), ``cli.py`` (argparse driver).

Why

The format is already well-defined and battle-tested — used by every model in ``exp0_models_protein_docs_initial_port/``, backing the ``timodonnell/protein-docs`` HF dataset's ``contacts-and-distances-v1-5x`` subset. Porting it is a mostly mechanical translation: doc-generation logic → ``generate.generate_documents``, in-training distogram-benchmark logic → ``inference.evaluate``, tokenizer construction → ``build_tokenizer(all_domain_tokens())``.

This is the first MarFold document-structure impl. It's the reference for any v2/v3 successors and the contract for downstream data-gen + eval experiments.

Document layout (reference)

```
`` <contacts-and-distances-v1> <begin_sequence> <AA_1> ... <AA_n> <begin_statements>  
<long-range-contact> <p_i> <p_j> <medium-range-contact> <p_i> <p_j> <distance> <p_i> <p_j>  
<atom_i> <atom_j> <d_value> <short-range-contact> <p_i> <p_j> <plddt_80_85> <end> ``
```

64 distance bins at 0.5 Å resolution; 3 contact modes by sequence separation; 7 pLDDT bins; position tokens ``-`<p2700>`; vocab 2840 tokens.

Success criteria

1. CLI subcommands ``generate / infer / evaluate / tokenizer`` parse and dispatch correctly (covered by ``tests/test_structure.py``). 2. Tokenizer byte-identical to the published ``timodonnell/protein-docs-tokenizer@83f597d88e9b``. 3. Tokenizing a generated 1QYS doc matches the token-ID sequence built by ``protein_distogram_eval.py`` today. 4. ``evaluate(cfg)`` on the 1B checkpoint reproduces the Phase 7c trend: zero-shot CA-CA MAE ~ 3.6 Å, dropping to ~ 1.4 Å with N=50 seeded GT contacts. 5. Graduates as ``document_structures/contacts_and_distances_v1/``.

Results so far

(Fill in as results come in.)

Conclusion

(Fill in after results are in.)